SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
```

```
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

ANSWERS:

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: Option B

Code:

```
<video src="video.mp4" controls></video>
```

Explanation:

The HTML5 <video> element is used to embed video files directly into a webpage. The src attribute specifies the path of the video file, while the controls attribute displays built-in playback controls such as play, pause, volume, and fullscreen.

Example:

```
<!DOCTYPE html>

<html>

<head>

   <title>Promotional Video</title>

</head>

<body>

   <h2>Our Promotional Video</h2>

   <video src="video.mp4" controls width="500">

      Your browser does not support video playback.

   </video>

</body>

</html>
```

Justification: Option B uses the standard HTML5 video element. The other options (<media>, <movie>, and <vid>) are not standard HTML elements for embedding videos.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: Option B — POST

Explanation:

The POST method is suitable for submitting registration form data to a web server. It sends the submitted data in the HTTP request body instead of placing it in the URL, as GET normally does.

Example:

```
<!DOCTYPE html>

<html>

<head>

   <title>Registration Form</title>

</head>

<body>

   <h2>Student Registration</h2>

   <form action="/register" method="POST">

      <label>Name:</label>

      <input type="text" name="username" required>

      <br><br>

      <label>Email:</label>

      <input type="email" name="email" required>

      <br><br>

      <label>Password:</label>

      <input type="password" name="password" required>

      <br><br>

      <button type="submit">Register</button>

   </form>

</body>

</html>
```

Justification: The method="POST" attribute specifies that the browser should submit the form data using HTTP POST. The server must process the request at the /register endpoint. In a real application, the endpoint should use HTTPS and handle passwords securely.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: Option B — <nav>

Explanation:

The <nav> element is a semantic HTML element used to define a major section of navigation links. It helps browsers, search engines, and assistive technologies identify the navigation area of a webpage.

Example:

```
<!DOCTYPE html>

<html>

<head>

  <title>Navigation Menu</title>

</head>

<body>

  <header>

    <h1>My Web Application</h1>

    <nav>

      <a href="index.html">Home</a>

      <a href="about.html">About</a>

      <a href="services.html">Services</a>

      <a href="contact.html">Contact</a>

    </nav>

  </header>

</body>

</html>
```

Justification: The <nav> element clearly identifies the navigation links and improves the semantic structure and accessibility of the webpage.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: Option B

Correct XML:

```
<?xml version="1.0" encoding="UTF-8"?>

<note>

  <to>User</to>

  <from>Admin</from>

</note>
```

Explanation:

XML (Extensible Markup Language) is used to store and transport structured data. XML requires properly nested elements, matching opening and closing tags, and exactly one root element.

Errors in the other options:

Option	Error
A	The <note> root element is not closed with </note>.
B	Correct syntax: all elements are properly opened, nested, and closed.
C	The <to> element opens but is incorrectly closed with </from>.
D	The <note> opening tag is not properly completed, and its closing tag is missing.

Justification: Option B is well-formed XML because every element has a matching closing tag, the elements are correctly nested, and <note> is the single root element.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: Option B — HTTP

Explanation:

HTTP stands for Hypertext Transfer Protocol. It is an application-layer protocol used for communication between web clients, such as browsers, and web servers. It follows a request-response model: the client sends an HTTP request, and the server returns an HTTP response.

Example of an HTTP request:

```
GET /index.html HTTP/1.1

Host: example.com
```

Example of an HTTP response:

```
HTTP/1.1 200 OK

Content-Type: text/html
```

The status code 200 OK indicates that the request was successful.

Justification: HTTP is designed for transferring webpages and related web resources. HTTPS adds TLS encryption and server authentication to protect web communication.

Therefore, HTTP is the correct choice for general web communication, while HTTPS is preferred for secure communication in modern applications.

Code of Conduct:

I Kandukuri Kavya (192425283) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Kandukuri Kavya

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand